

Syllabus: ECON-GA 1802, Industrial Organization II, Spring 2025

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Office Hours : By appointment
Lecture : Monday, 10:30am – 12:30pm, 19 W 4th Room 624

Description

The class covers recent developments in the empirical analysis of theoretical models of market equilibrium and market power. In combination with ECON-GA 1801 and Chris Conlon's empirical IO course at Stern (ECON-GB 3360), it will prepare you to write a dissertation in empirical industrial organization. The first part of the course covers methods in the empirical IO toolkit, including static and dynamic models of entry, incomplete models, dynamic games, and auctions. The second part of the course applies these methods to topics such as market design, health care, and transportation. Throughout, the focus is on how to explicitly combine data and theory to answer an economic question of interest.

Lecture Schedule

Jan 27	Entry Models	
Feb 3	Entry and Multiple Equilibria	
Feb 10	Dynamic Games I: Methods	
<i>Feb 18</i>	Dynamic Games II: Applications	<i>[note: legislative Monday]</i>
Feb 24	Auctions	
Mar 3	Market Design I (static)	
Mar 10	Market Design II (static)	
Mar 17	Market Design III (dynamics)	
Mar 24	SPRING BREAK	
Mar 31	Selection Markets I: Methods	
Apr 7	Selection Markets II: Applications	
Apr 14	TBD	
Apr 21	TBD	
Apr 28	TBD	
May 5	TBD	

Pre-requisites and required textbooks

This course assumes students have taken Chris Conlon's "Topics: Industrial Organization" course at Stern or the equivalent, as well as Jovanovic's course (IO I) in the economics department. Most importantly, students should be familiar with the IO literature on static and dynamic demand estimation (BLP, Rust, and Hotz-Miller) and have experience coding structural econometric models. Knowledge of the material from the first-year PhD econometrics sequence is also needed to understand the readings and complete assignments. Students are expected to pick up additional econometric methods as necessary.

There is no required textbook. Students interested in writing a thesis in IO will benefit from owning a copy of *The Theory of Industrial Organization* by Jean Tirole.

Class Requirements

Assignments (50%): There will be two long problem sets. You will derive estimators and conduct empirical and counterfactual exercises that are simplified versions of problems one is likely to encounter when writing a paper in empirical IO.

You must submit your own original code and write-up for each assignment. There is simply no substitute for deriving and coding these estimators yourself. That said, you are welcome and encouraged to discuss the assignments with your classmates. Please list the all the students with whom you have discussed the assignment on your problem set submission. The answers are due via the class website. Please typeset your answers and turn in PDF files and your source code.

While I will accept code in any language, you are encouraged to use common programming languages such as MATLAB, R, Julia, or Python (MATLAB is easiest for me). Please write code that is well-documented, follows transparent logic, and uses short, descriptive variable names. Most importantly, ensure that your results can be easily reproduced. I should be able to run your code by calling a main script file that produces any tables, figures, or numbers used in your write-up. If I cannot reproduce your results, you will not receive credit for the assignment.

Reading/Participation (25%): Each week, I will designate one paper that you are expected to read in detail and submit a written response by midnight (EST) the evening before lecture. These written responses should be short – 1-2 pages double-spaced – and should summarize the paper that will be discussed in class, as well as identify any flaws in the paper or questions that you may have. Lecture will focus on this paper and one or two others, which you should also look at before class. Response papers are marked with three stars on the syllabus, other required papers with one star.

Referee Report (25%): During the second half of the course, you will be assigned a referee report on a recent paper that exemplifies the empirical methods and applications covered in lecture. See the referee report guidelines on the class website for specific instructions. You will be given at least two weeks to complete the report.

Assignment Schedule

Feb 24	Problem Set 1
Mar 24	Problem Set 2
Apr 6	Referee Report

Reading List

Handbook Chapters

Akerberg, D., Benkard, C., Berry, S., & Pakes, A. (2007). Econometric Tools for Analyzing Market Outcomes. *Handbook of Econometrics*, 6, 4171-4276.

Agarwal, N., Budish, E. (2021). Market design. *Handbook of Industrial Organization*, 5, 1-79.

Aguirregabiria, V., Collard-Wexler, A., & Ryan, S. P. (2021). Dynamic games in empirical industrial organization. *Handbook of Industrial Organization*, 4, 225-343.

Athey, S., & Haile, P. A. (2007). Nonparametric Approaches to Auctions. *Handbook of Econometrics*, 6, 3847-3965.

Berry, S., & Reiss, P. (2007). Empirical Models of Entry and Market Structure. *Handbook of Industrial Organization*, 3, 1845-1886.

Bresnahan, T. F. (1989). Empirical Studies of Industries with Market Power. *Handbook of Industrial Organization*, 2, 1011-1057.

Einav, L., Finkelstein, A., & Mahoney, N. (2021). The IO of selection markets. *Handbook of Industrial Organization*, 4, 389-426.

Hendricks, K., & Porter, R. H. (2007). An Empirical Perspective on Auctions. *Handbook of Industrial Organization*, 3, 2073-2143.

Hortacsu, A. & Perrigne, I. (2021). Empirical perspectives on auctions. *Handbook of Industrial Organization*, 5, 81-175.

Matzkin, R. L. (2007). Nonparametric Identification. *Handbook of Econometrics*, 6, 5307-5368.

Newey, W. K., & McFadden, D. (1994). Large Sample Estimation and Hypothesis Testing. *Handbook of Econometrics*, 4, 2111-2245.

Reiss, P. C., & Wolak, F. A. (2007). Structural Econometric Modeling: Rationales and Examples from Industrial Organization. *Handbook of Econometrics*, 6, 4277-4415.

Methodological Readings

Agarwal, N. & P. Somaini (2020). Revealed Preference Analysis of School Choice Models. *Annual Review of Economics*, 12, 471-501.

Chernozhukov, V., Hong, H., & Tamer, E. (2007). Estimation and Confidence Regions for Parameter Sets in Econometric Models. *Econometrica*, 75(5), 1243-1284.

Gelman, A., Carlin, J.B., Stern, H.S., Dunson, D.B., Vehtari, A., and D.B. Rubin (2013). *Bayesian Data Analysis*, Third Edition. CRC Press.

McFadden, D. (1989). A Method of Simulated Moments for Estimation of Discrete Response Models without Numerical Integration. *Econometrica*, 995-1026.

Pakes, A. "Alternative models for moment inequalities." *Econometrica* 78.6 (2010): 1783-1822.

Pakes, A., & Pollard, D. (1989). Simulation and the asymptotics of optimization estimators. *Econometrica: Journal of the Econometric Society*, 1027-1057.

Pakes, A., Porter, J., Ho, K., & Ishii, J. (2015). Moment inequalities and their application. *Econometrica*, 83(1), 315-334.

Romano, J. P., & Shaikh, A. M. (2010). Inference for the identified set in partially identified econometric models. *Econometrica*, 78(1), 169-211.

Entry

Baker, J. B., & Bresnahan, T. F. (1985). The gains from merger or collusion in product-differentiated industries. *The Journal of Industrial Economics*, 427-444.

***Berry, S. T. (1992). Estimation of a Model of Entry in the Airline Industry. *Econometrica: Journal of the Econometric Society*, 889-917.

Berry, S. T., & Waldfogel, J. (1999). Free Entry and Social Inefficiency in Radio Broadcasting. *The Rand Journal of Economics*, 397-420.

*Bresnahan, T. F., & Reiss, P. C. (1991). Entry and competition in concentrated markets. *Journal of Political Economy*, 977-1009.

Fan, Y. (2013). Ownership consolidation and product characteristics: A study of the US daily newspaper market. *The American Economic Review*, 103(5), 1598-1628.

Gowrisankaran, G. (1999). A dynamic model of endogenous horizontal mergers. *The RAND Journal of Economics*, 56-83.

Hsieh, C. T., & Moretti, E. (2003). Can Free Entry Be Inefficient? Fixed Commissions and Social Waste in the Real Estate Industry. *Journal of Political Economy*, 111(5), 1076-1122.

Jia, P. (2008). What Happens When Wal-Mart Comes to Town: An Empirical Analysis of the Discount Retailing Industry. *Econometrica*, 76(6), 1263-1316.

Mazzeo, M. J. (2002). Product choice and oligopoly market structure. *RAND Journal of Economics*, 221-242.

Mankiw, N. G., & Whinston, M. D. (1986). Free entry and social inefficiency. *The RAND Journal of Economics*, 48-58.

Nevo, A. (2000). Mergers with Differentiated Products: The Case of the Ready-to-Eat Cereal Industry. *RAND Journal of Economics*, 31(3), 395-421.

Nosko, C. (2010). Competition and quality choice in the CPU market. *Manuscript, Harvard University*.

*Seim, K. (2006). An empirical model of firm entry with endogenous product-type choices. *The RAND Journal of Economics*, 37(3), 619-640.

Shapiro, C. (2010). The 2010 horizontal merger guidelines: From Hedgehog to Fox in forty years. *Antitrust Law Journal*, 49-107.

Sweeting, A. (2009). The strategic timing incentives of commercial radio stations: An empirical analysis using multiple equilibria. *The RAND Journal of Economics*, 40(4), 710-742.

*Wollman, T.G. (2018). Trucks without Bailouts: Equilibrium Product Characteristics for Commercial Vehicles. *American Economic Review*, 108(6), 1364-1406.

Moment Inequality Models

Andrews, D. W. K., & G. Soares (2010). Inference for Parameters Defined by Moment Inequalities using Generalized Moment Selection. *Econometrica*, 78(1), 119-157.

*Chernozhukov, V., Hong, H., & Tamer, E. (2007). Estimation and Confidence Regions for Parameter Sets in Econometric Models. *Econometrica*, 75(5), 1243-1284.

***Ciliberto, F. & E. Tamer (2009). Market Structure and Multiple Equilibria in Airline Markets. *Econometrica*, 77(6), 1791-1828.

Dickstein, M., & E. Morales (2018). What Do Exporters Know? *The Quarterly Journal of Economics*, 133(4), 1753-1801.

Galichon, A., & M. Henry (2011). Set Identification in Models with Multiple Equilibria. *Review of Economic Studies*, 78(2), 1264-1298.

***Illanes, G. (2017). Switching Costs in Pension Plan Choice. *Manuscript, Northwestern*.

Jovanovic, B. (1989). Observable Implications of Models with Multiple Equilibria. *Econometrica*, 57(6), 1431-1437.

*Pakes A (2010). [Alternative Models for Moment Inequalities](#). *Econometrica*, 78(6), 1783-1822.

Pakes, A., & J. Porter (2014). Moment Inequalities for Multinomial Choice with Fixed Effects. *Manuscript, Harvard University*.

*Pakes, A., Porter, J., Ho, K., Ishii, J. (2015). Moment Inequalities and their Application. *Econometrica*, 83(1), 315-334.

Schennach, S. M. (2014). Entropic Latent Variable Integration via Simulation. *Econometrica*, 82(1), 345-385.

Dynamics

Abbring, J. H. & J. R. Campbell (2010). Last-In First-Out Oligopoly Dynamics. *Econometrica*, 78 (5): 1491–1527.

Aguirregabiria, V. and Mira, P. (2002), Swapping the Nested Fixed Point Algorithm: A Class of Estimators for Discrete Markov Decision Models. *Econometrica*, 70: 1519–1543.

*Aguirregabiria, V., & Mira, P. (2007). Sequential Estimation of Dynamic Discrete Games. *Econometrica*, 75(1), 1-53.

Aguirregabiria, V., & A. Nevo. (2013). Recent Developments in Empirical IO: Dynamic Demand and Dynamic Games, in D. Acemoglu, M. Arellano, and E. Deckel, eds., *Advances in Economics and Econometrics. Tenth World Congress of the Econometric Society. Volume 3, Econometrics*.

See also: Textbook-like IO course materials by Victor Aguirregabiria (U Toronto):
http://individual.utoronto.ca/vaguirre/courses/eco2901/teaching_io_toronto.html

Arcidiacono, Peter, Patrick Bayer, Jason R. Blevins, Paul B. Ellickson. (2016). Estimation of Dynamic Discrete Choice Models in Continuous Time with an Application to Retail Competition, *Review of Economic Studies*, 83(3), 889–931.

Arcidiacono, P., & Miller, R. A. (2011). Conditional Choice Probability Estimation of Dynamic Discrete Choice Models with Unobserved Heterogeneity. *Econometrica*, 79(6), 1823-1867.

***Bajari, P., Benkard, C. L., & Levin, J. (2007). Estimating Dynamic Models of Imperfect Competition. *Econometrica*, 75(5), 1331-1370.

Benkard, C. L. (2000). Learning and Forgetting: The Dynamics of Aircraft Production. *American Economic Review*, 90(4), 1034-1054.

Benkard, C. L. (2004). A Dynamic Analysis of the Market for Wide-Bodied Commercial Aircraft. *Review of Economic Studies*, 71 (3): 581–611.

Benkard, C. L., Jeziorski, P., & Weintraub, G. Y. (2015). Oblivious equilibrium for concentrated industries. *RAND Journal of Economics*, 46(4), 671-708.

Berry, S. T., & G. Compiani. (2021). An Instrumental Variable Approach to Dynamic Models. *Review of Economic Studies* (forthcoming).

Besanko, D., Doraszelski, U., Kryukov, Y. and Satterthwaite, M. (2010), Learning-by-Doing, Organizational Forgetting, and Industry Dynamics. *Econometrica*, 78: 453–508.

Buchholz, N., M. Shum, & H. Xu. (2021). Semiparametric Estimation of Dynamic Discrete Choice Models. *Journal of Econometrics*, 223, 312-327.

*Collard-Wexler, A. (2013). Demand Fluctuations in the Ready-Mix Concrete Industry. *Econometrica*, 81(3), 1003-1037.

See also: Well-structured IO (“Entry-model”) course materials by Allan Collard-Wexler (Duke): https://sites.duke.edu/collardwexler/teaching/econ_881-15/

Dixit, A. (1989). Entry and Exit Decisions under Uncertainty. *Journal of Political Economy*, 620-638.

Doraszelski, U. and Judd, K. L. (2012), Avoiding the Curse of Dimensionality in Dynamic Stochastic Games. *Quantitative Economics*, 3, 53–93.

Doraszelski, Ulrich, and Mark Satterthwaite (2010). Computable Markov-Perfect Industry Dynamics. *RAND Journal of Economics*, 41 (2): 215-243.

Egedal, Michael, Zhenyu Lai, and Che-Lin Su (2015). Estimating Dynamic Discrete-Choice Games of Incomplete Information. *Quantitative Economics*, 6: 567-597.

Ericson, R., & Pakes, A. (1995). Markov-Perfect Industry Dynamics: A Framework for Empirical Work. *The Review of Economic Studies*, 62(1), 53-82.

Fershtman, C., & Pakes, A. (2000). A Dynamic Oligopoly with Collusion and Price Wars. *RAND Journal of Economics*, 31(2), 207-236.

*Fershtman C, & Pakes A. (2012). Dynamic Games with Asymmetric Information: A Framework for Empirical Work. *Quarterly Journal of Economics*, 127(4):1611-1661.

Goettler, Ronald, and Brett Gordon (2011). Does AMD Spur Intel to Innovate More? *Journal of Political Economy*, 119(6): 1141-1200.

Gowrisankaran, Gautam. (1999). A Dynamic Model of Endogenous Horizontal Mergers. *RAND Journal of Economics*, Vol. 30, 56-83.

Gowrisankaran, G., & Rysman, M. (2012). Dynamics of Consumer Demand for New Durable Goods. *Journal of Political Economy*, 120(6), 1173-1219.

Gowrisankaran, G., & Town, R. J. (1997). Dynamic Equilibrium in the Hospital Industry. *Journal of Economics & Management Strategy*, 6(1), 45-74.

Hendel, I., & Nevo, A. (2006). Measuring the Implications of Sales and Consumer Inventory Behavior. *Econometrica*, 74(6), 1637-1673.

Holmes, T. J. (2011). The Diffusion of Wal-Mart and Economies of Density. *Econometrica*, 79 (1): 253–302.

Hopenhayn, H. A. (1992). Entry, Exit, and Firm Dynamics in Long Run Equilibrium. *Econometrica*, 60 (5): 1127-1150.

Hotz, V. J., & Miller, R. A. (1993). Conditional Choice Probabilities and the Estimation of Dynamic Models. *The Review of Economic Studies*, 60(3), 497-529.

Hotz, V. Joseph, Robert A. Miller, Seth Sanders, Jeffrey Smith. A Simulation Estimator for Dynamic Models of Discrete Choice, *The Review of Economic Studies*, 61 (2): 265–289.

Igami, M. (2017). Estimating the Innovator's Dilemma: Structural Analysis of Creative Destruction in the Hard Disk Drive Industry, 1981–1998. *Journal of Political Economy*, 125 (3): 798–847.

Igami, M. (2020). Artificial Intelligence as Structural Estimation: Deep Blue, Bonanza, and AlphaGo. *The Econometrics Journal*, 23(3), S1-S24.

Igami, M. (2018). Industry Dynamics of Offshoring: The Case of Hard Disk Drives. *American Economic Journal: Microeconomics*, 10(1), 67-101.

Igami, M. & T. Sugaya (2021). Measuring the Incentive to Collude: The Vitamin Cartels, 1990–1999. *Review of Economic Studies* (forthcoming).

***Igami, M. & K. Uetake (2020). Mergers, Innovation, and Entry-Exit Dynamics: Consolidation of the Hard Disk Drive Industry, 1996–2016. *The Review of Economic Studies*, 87(6), 2672-2702.

Igami, M. & N. Yang (2016). Unobserved Heterogeneity in Dynamic Games: Cannibalization and Preemptive Entry of Hamburger Chains in Canada. *Quantitative Economics*, 7 (2): 483–521.

Iskhakov, F., J. Rust, and B. Schjerning (2018). The Dynamics of Bertrand Price Competition with Cost-Reducing Investments. *International Economic Review*, 59(4), 1681-1731.

Iskhakov, F., J. Rust, and B. Schjerning (2016). Recursive Lexicographical Search: Finding All Markov Perfect Equilibria of Finite State Directional Dynamic Games. *Review of Economic Studies*, 83(2), 658-703.

Jeziorski, Przemyslaw (2015). Empirical Model of Dynamic Merger Enforcement: Choosing Ownership Caps in U.S. Radio. *Manuscript, UC Berkeley*.

Jofre-Bonet, M., & Pesendorfer, M. (2003). Estimation of a Dynamic Auction Game. *Econometrica*, 71(5), 1443-1489.

- Jovanovic, B. (1982). Selection and the Evolution of Industry. *Econometrica*, 50 (3): 649-670.
- Kalouptside, M. (2014). Time to Build and Fluctuations in Bulk Shipping. *The American Economic Review*, 104(2), 564-608.
- Kalouptside, M., P. T. Scott, E. Souza-Rodrigues (2020). Identification of Counterfactuals in Dynamic Discrete Choice Models. *Quantitative Economics*, 20(2).
- Kasahara, H., & Shimotsu, K. (2009). Nonparametric Identification of Finite Mixture Models of Dynamic Discrete Choices. *Econometrica*, 77(1), 135-175.
- Marshall, G., & A. Parra. (2019). Innovation and Competition: The Role of the Product Market. *International Journal of Industrial Organization*, 65, 221-247.
- Maskin, E., & Tirole, J. (1988). A Theory of Dynamic Oligopoly, II: Price Competition, Kinked Demand Curves, and Edgeworth Cycles. *Econometrica*, 571-599.
- Mazur, J. (2017). Can Stricter Bankruptcy Laws Discipline Capital Investment? Evidence from the U.S. Airline Industry. *Manuscript, Purdue University*.
- Pakes, A. (1986). Patents as Options: Some Estimates of the Value of Holding European Patent Stocks. *Econometrica*, 54(4), 755-784.
- Pakes, A., & McGuire, P. (1994). Computing Markov-Perfect Nash Equilibria: Numerical Implications of a Dynamic Differentiated Product Model. *RAND Journal of Economics*, 25(4), 555-589.
- *Pakes, A., Ostrovsky, M., & Berry, S. (2007). Simple Estimators for the Parameters of Discrete Dynamic Games (with Entry/Exit Examples). *The RAND Journal of Economics*, 38(2), 373-399.
- Pesendorfer, Martin, and Philipp Schmidt-Dengler (2008). Asymptotic Least Squares Estimators for Dynamic Games. *Review of Economic Studies*, 75 (3): 901--928.
- Rust, J. (1987). Optimal replacement of GMC bus engines: An empirical model of Harold Zurcher. *Econometrica: Journal of the Econometric Society*, 999-1033.
- Rust, J. (2017). Dynamic Programming, Numerical. *Wiley Stats Ref: Statistics Reference Online*.
- ***Ryan, S. P. (2012). The Costs of Environmental Regulation in a Concentrated Industry. *Econometrica*, 80(3), 1019-1061.
- Scott, P. T. (2013). Dynamic Discrete Choice Estimation of Agricultural Land Use. *Manuscript, Toulouse School of Economics*.
- Su, C.L. and Judd, K.L. (2012). Constrained Optimization Approaches to Estimation of Structural Models. *Econometrica*, 80: 2213–2230.

*Sweeting, A. (2013). Dynamic Product Positioning in Differentiated Product Markets: The Effect of Fees for Musical Performance Rights on the Commercial Radio Industry. *Econometrica*, 81(5), 1763-1803.

Takahashi, Yuya (2015). Estimating a War of Attrition: The Case of the US Movie Theater Industry. *American Economic Review*, 105 (7): 2204–41.

*Weintraub, G. Y., Benkard, C. L. and Van Roy, B. (2008). Markov Perfect Industry Dynamics with Many Firms. *Econometrica*, 76: 1375–1411.

Weintraub, G. Y., Benkard, C. L., & Van Roy, B. (2010). Computational Methods for Oblivious Equilibrium. *Operations Research*, 58, 1247-1265.

Williams, K. R. (2017). Dynamic Airline Pricing and Seat Availability. *Cowles Foundation Discussion Paper No. 2103*.

Yang, Chenyu. (2017). Vertical Structure and Innovation: A Study of the SoC and Smartphone Industries. *RAND Journal of Economics*, 51(3), 739-785.

Zheng, Fanyin. (2016). Spatial Competition and Preemptive Entry in the Discount Retail Industry. *Manuscript, Columbia University*.

Auctions

Aradillas-López, A., Gandhi, A., & Quint, D. (2013). Identification and Inference in Ascending Auctions with Correlated Private Values. *Econometrica*, 81(2), 489-534.

Asker, J. (2010). A Study of the Internal Organization of a Bidding Cartel. *The American Economic Review*, 724-762.

*Athey, S., & Haile, P. A. (2002). Identification of Standard Auction Models. *Econometrica*, 70(6), 2107-2140.

Athey, S., & Levin, J. (2001). Information and Competition in US Forest Service Timber Auctions. *Journal of Political Economy*, 109(2).

Athey, S., Levin, J., & Seira, E. (2011). Comparing Open and Sealed Bid Auctions: Evidence from Timber Auctions. *The Quarterly Journal of Economics*, 126(1), 207-257.

Bajari, P., & Fox, J. T. (2013). Measuring the Efficiency of an FCC Spectrum Auction. *American Economic Journal: Microeconomics*, 5(1), 100-146.

Bhattacharya, V. (2020). An Empirical Model of R&D Procurement Contests: An Analysis of the DOD SBIR Program. *Econometrica*, 89(5), 2189-2224.

*Bolotnyy, V. & S. Vasserman. Scaling Auctions as Insurance: A Case Study in Infrastructure Procurement. *Manuscript, Stanford University*.

Campo, S., Perrigne, I., & Vuong, Q. (2003). Asymmetry in First-Price Auctions with Affiliated Private Values. *Journal of Applied Econometrics*, 18(2), 179-207.

Cassola, N., Hortaçsu, A., & Kastl, J. (2013). The 2007 Subprime Market Crisis Through the Lens of European Central Bank Auctions for Short-Term Funds. *Econometrica*, 81(4), 1309-1345.

Chassang, S., Kawai, K., Nakabayashi, J., & J. Ortner (2019). Data Driven Regulation: Theory and Application to Missing Bids. *Manuscript*.

De los Santos, B., Hortacsu, A., & Wildenbeest, M.R. (2012). Testing Models of Consumer Search Using Data on Web Browsing and Purchasing Behavior. *American Economic Review*, 102(6), 2955-2980.

Donna, J.D. & J.A. Espin-Sanchez (2018). The Illiquidity of Water Markets: Efficient Institutions for Water Allocation in Southeastern Spain. *Manuscript*.

***Guerre, E., Perrigne, I., & Vuong, Q. (2000). Optimal Nonparametric Estimation of First-Price Auctions. *Econometrica*, 68(3), 525-574.

Haile, P. A., Hong, H., & Shum, M. (2003). *Nonparametric Tests for Common Values at First-Price Sealed-Bid Auctions*. National Bureau of Economic Research, Working Paper No. w10105.

*Haile, P. A., & Tamer, E. (2003). Inference with an Incomplete Model of English Auctions. *Journal of Political Economy*, 111(1), 1-51.

Hortacsu, A. & Bajari, P. (2005). Are Structural Estimates of Auction Models Reasonable? Evidence from Experimental Data. *Journal of Political Economy*, 113(4), 703-741.

Larsen, B. (2020). The Efficiency of Real-World Bargaining: Evidence from Wholesale Used-Auto Auctions. *Review of Economic Studies* (forthcoming).

Kastl, J. (2011). Discrete Bids and Empirical Inference in Divisible Good Auctions. *The Review of Economic Studies*, 78(3), 974-1014.

Krasnokutskaya, E. (2011). Identification and Estimation of Auction Models with Unobserved Heterogeneity. *The Review of Economic Studies*, 78(1), 293-327.

Laffont, J. J., Ossard, H., & Vuong, Q. (1995). Econometrics of First-Price Auctions. *Econometrica*, 953-980.

Li, T., Perrigne, I., & Vuong, Q. (2002). Structural Estimation of the Affiliated Private Value Auction Model. *RAND Journal of Economics*, 171-193.

Porter, R. H., & Zona, J. D. (1993). Detection of Bid Rigging in Procurement Auctions. *Journal of Political Economy*, 101(3), 518-38.

Reguant, M. (2014). Complementary Bidding Mechanisms and Start-Up Costs in Electricity Markets. *Review of Economic Studies*, 81(4), 1708-1742.

Roberts, J. W. (2013). Unobserved Heterogeneity and Reserve Prices in Auctions. *The RAND Journal of Economics*, 44(4), 712-732.

Slattery, C. (2019). Bidding for Firms: Subsidy Competition in the U.S. *Manuscript*.

*Somaini, P. (2020). Identification in Auction Models with Interdependent Costs. *Journal of Political Economy*, 128(10), 3820-3871.

Market Design and Matching

***Agarwal, N., Ashlagi, I., Rees, M., Somaini, P., & D. Waldinger (2020). Equilibrium Allocations under Alternative Waitlist Designs: Evidence from Deceased Donor Kidneys. *Econometrica*, 89(1), 37-76.

Abdulkadiroğlu, A., Agarwal, N., & Pathak, P. A. (2017). The Welfare Effects of Coordinated Assignment: Evidence from the New York City High School Match. *The American Economic Review*, 107(12), 3635-3689.

*Agarwal, N. (2015). An Empirical Model of the Medical Match. *The American Economic Review*, 105(7), 1939-78.

***Agarwal, N., & P. Somaini. (2018). Demand Analysis using Strategic Reports: An Application to a School Choice Mechanism. *Econometrica*, 86(2), 391-444.

Boyd, D., Lankford, H., Loeb, S., & Wyckoff, J. (2013). Analyzing the Determinants of the Matching of Public School Teachers to Jobs: Disentangling the Preferences of Teachers and Employers. *Journal of Labor Economics*, 31(1), 83-117.

Bucarey, A. (2018). Who Pays for Free College? Crowding Out on Campus. *Manuscript, MIT*.

*Calsamiglia, C., Fu, C., & Güell, M. (2019). Structural Estimation of a Model of School Choices: the Boston Mechanism vs. Its Alternatives. *Journal of Political Economy*, 128(2), 642-680.

Chiappori, P. A., Salanié, B., & Weiss, Y. (2017). Partner Choice, Investment in Children, and the Marital College Premium. *American Economic Review*, 107(8), 2109-67.

Choo, E., & Siow, A. (2006). Who Marries Whom and Why. *Journal of Political Economy*, 114(1), 175-201.

Diamond, W. and Agarwal, N. (2017). Latent Indices in Assortative Matching Models. *Quantitative Economics*, 8(3), 685-728.

- *Fack, G., Grenet, J., & He, Y. (2019). Beyond Truth-Telling: Preference Estimation with Centralized School Choice and College Admission. *American Economic Review*, 109(4), 1486-1529.
- Fox, J. T. (2008). *Estimating matching games with transfers* (No. w14382). National Bureau of Economic Research.
- Fox, J. T. (2010). Identification in Matching Games. *Quantitative Economics*, 1(2), 203-254.
- Gale, D., & Shapley, L. S. (1962). College Admissions and the Stability of Marriage. *American Mathematical Monthly*, 9-15.
- Galichon, A., & Salanié, B. (2010). Matching with Trade-offs: Revealed Preferences over Competing Characteristics. *CEPR Discussion Paper No. DP7858*.
- Hastings, J. S., Kane, T. J., & Staiger, D. (2005). Parental Preferences and School Competition: Evidence from a Public School Choice Program. *NBER Working Paper*, (w11805).
- He, Y. (2017). Gaming the Boston School Choice Mechanism in Beijing. *Manuscript, Rice University*.
- *Kapor, A., Neilsen, C., & Zimmerman, S. (2020). Heterogeneous Beliefs and School Choice. *American Economic Review*, 110(5), 1274-1315.
- Menzel, K. (2015). Large Matching Markets as Two-Sided Demand Systems. *Econometrica*, 83(3), 897-941.
- Reeling, C. & V. Verdier (2020). Welfare Effects of Dynamic Matching: An Empirical Analysis. *Review of Economic Studies* (forthcoming).
- Roth, A. E., & Sotomayor, M. A. O. (1992). Two-sided matching: A study in game-theoretic modeling and analysis (No. 18). *Cambridge University Press*.
- Sørensen, M. (2007). How smart is smart money? A two-sided matching model of Venture Capital. *The Journal of Finance*, 62(6), 2725-2762.
- Uetake, K., & Watanabe, Y. (2012). Entry by Merger: Estimates from a Two-Sided Matching Model with Externalities. *Available at SSRN 2188581*.
- *Waldinger, D. (2020). Targeting In-Kind Transfers through Market Design: A Revealed Preference Analysis of Public Housing Allocation. *American Economic Review*, 111(8), 2660-2696.

Selection Markets

- *Cohen, A. & L. Einav (2007). Estimating Risk Preferences from Deductible Choice. *American Economic Review*, 97(3), 745-788.

Cuesta, J. & A. Sepulveda (2019). Price Regulation in Credit Markets: A Trade-off between Consumer Protection and Credit Access. *Manuscript*.

Decarolis, F., Polyakova, M., & S. Ryan (2020). Subsidy Design in Privately-Provided Social Insurance: Lessons from Medicare Part D. *Journal of Political Economy*, 128(5).

Einav, L., Finkelstein, F., & P. Schrimpf (2010). Optimal Mandates and the Welfare Cost of Asymmetric Information: Evidence from the U.K. Annuity Market. *Econometrica*, 78(3), 1031-1092.

*Einav, L., Finkelstein, F., Schrimpf, P., Ryan, S., & M.R. Cullen (2013). Selection on Moral Hazard in Health Insurance. *American Economic Review*, 103(1), 178-219.

Handel, B. (2013). Adverse Selection and Inertia in Health Insurance Markets: When Nudging Hurts. *American Economic Review*, 103(7), 2643-2682.

Handel, B., Hendel, I., & M.D. Whinston (2015). Equilibria in Health Exchanges: Adverse Selection versus Reclassification Risk. *Econometrica*, 83(4), 1261-1313.

Ho, K. (2009). Insurer-Provider Networks in the Medical Care Market. *American Economic Review*, 99(1), 393-430.

Jin, Y. & Vasserman, S. (2020). Buying Data from Consumers: the Impact of Monitoring Programs in U.S. Auto Insurance. *Manuscript, UC Berkeley*.

*Marone, V. (2020). Should there be Vertical Choice in Health Insurance Markets? *Manuscript, Northwestern*.

Mahoney, N. & E. G. Weyl (2017). Imperfect Competition in Selection Markets. *Review of Economics and Statistics*, 99(4), 637-651.

***Nelson, S. (2020). Private Information and Price Regulation in the US Credit Card Market. *Manuscript, University of Chicago*.

Shepard, M. (2016). Hospital Network Competition and Adverse Selection: Evidence from the Massachusetts Health Insurance Exchange. *Manuscript, Harvard*.

*Tiebaldi, P., Torgovitsky, A., & H. Yang (2021). Nonparametric Estimates of Demand in the California Health Insurance Exchange. *Manuscript, Columbia University*.

*Vatter, B. (2021). Quality Disclosure and Regulation: Scoring Design in Medicare Advantage. *Manuscript, Northwestern*.

Schnell, M. (2017). Physician Behavior in the Presence of a Secondary Market: The Case of Prescription Opioids. *Manuscript, Princeton University*.

IO Search, Transportation

- Allen, J., Clark, R., & J.F. Houde (2013). The Effect of Mergers in Search Markets: Evidence from the Canadian Mortgage Industry. *American Economic Review*, 104(10), 3365-3396.
- Brancaccio, G., Kalouptsi, M., & T. Papageorgiou (2020). Geography, Transportation and Endogenous Trade Costs. *Econometrica*, 88(2), 657-691.
- Brancaccio, G., Li, D., & N. Schurhoff (2021). Learning by Trading: The Case of the U.S. Market for Municipal Bonds. *Mimeo, Cornell University*.
- Buchholz, N. (2021). Spatial Equilibrium, Search Frictions and Dynamic Efficiency in the Taxi Industry. *Review of Economic Studies (forthcoming)*.
- Buchholz, N., Doval, L., Kastl, J., Matejka, F., & Salz, T. (2020). The Value of Time: Evidence from Auctioned Cab Rides. *Manuscript, Princeton University*.
- Castillo, J.C. (2020). Who Benefits from Surge Pricing? *Manuscript, University of Pennsylvania*.
- Coey, D., Larsen, B., & B.C. Platt (2020). Discounts and Deadlines in Consumer Search. *American Economic Review*, 110(12), 3748-3786.
- De los Santos, B., Hortacsu, A., & M. Wildenbeest (2012). Testing Models of Consumer Search using Data on Web Browsing and Purchasing Behavior. *American Economic Review*, 102(6), 2955-80.
- Frechette, G., Lizzeri, A., & T. Salz (2019). Frictions in a Competitive, Regulated Market: Evidence from Taxis. *American Economic Review*, 109(8), 2954-2992.
- Hortacsu, A. & C. Syverson (2004). Product Differentiation, Search Costs and Competition in the Mutual Fund Industry: A Case Study of S&P 500 Index Funds. *Quarterly Journal of Economics*, 119, 403-456.
- Liu, T.X., Wan, Z., & C. Yang (2019). The Efficiency of a Dynamic Decentralized Two-sided Matching Market. *Manuscript, University of Rochester*.
- Salz, T. (2022). Intermediation and Competition in Search Markets: An Empirical Case Study. *Journal of Political Economy*, 130(1).